

COM S 327, Spring 2026

Programming Project 1.08

Loading Pokémon

We've parsed in a number of data files that specify certain details about how to create pokémon. Now we're actually going to load them into our game. We'll add the ability to encounter them, but we'll save battling and capturing for next week.

We're going to simplify many of the mechanics of the pokémon *main series games* (MSGs) in our implementation. The full mechanics are just too tedious, which would result in a lot of code and not much learning. If faithful (or more faithful) mechanics are important to you, you're welcome to implement something more detailed than presented here.

When walking through tall grass, each move the PC makes has a certain chance of encountering a pokémon. In the MSGs, different areas of the world will have different kinds of pokémon in the grass, as well as other types of terrain (desert, caves, water) that may spawn pokémon, as well. We'll simplify this by allowing any (literally any!) pokémon to spawn in any tall grass, this includes fully evolved forms and legendary and mythical pokémon. We've parsed in details on 1092 pokémon; we'll select from them on a uniform distribution, which means that you have an equal chance of encountering a pidgey and a mewtwo. As for the probability of encountering a pokémon in the tall grass, something like 10% seems reasonable, but you're welcome to play with the number.

The levels of encountered pokémon will increase with the manhattan distance from the origin. When distance is less than or equal to 200, minimum level is 1 and maximum level is distance / 2 (integer division). When distance exceeds 200, minimum level becomes (distance - 200) / 2 and maximum level is 100. With this formulation, at the origin, all encountered pokémon are level 1, if you go all the way east without going north or south, pokémon are generated at all possible levels, and if you then go all the way north you'll encounter only level 100 pokémon.

Encountered pokémon will know up to two moves. The known moves are restricted to the pokémon's *level-up learnset* and will only be less than two if the pokémon has fewer than two moves in its level-up learnset at its level. If a pokémon does not have at least one move in its level-up learnset, advance its level to the point where it can learn a move¹.

The pokémon in our database are grouped in many ways. The `version_group_id` in `pokemon_moves.csv` is a unique identifier of a pokémon game. Theoretically, each game contains all of the pokémon from all of the earlier games, however that ended ended with *Pokémon Sword and Shield* in a controversial event that came to be known as *Dexit*². What this means to us is that there exists no single game from which we can pull movesets and expect that all pokémon will have a non-empty set. If you want your pokémon to have movesets from a specific game, you may search for a particular `version_group_id`; otherwise, ignore this field.

To find the level-up moveset of a pokémon, you'll want `pokemon_moves.pokemon_id` equal to `pokemon.species_id` and `pokemon_moves.pokemon_move_method_id` equal to 1. From these, select your two moves from a uniform distribution. Given the pokémon's level-up moveset, you then select up to two moves from that set where `pokemon_moves.level` is less than or equal to the pokémon's level and assign those moves to your pokémon.

¹Or give it *Struggle*.

²A portmanteau of *Pokédex* and *Brexit*; <https://www.newsweek.com/pokemon-sword-shield-dexit-explained-why-fans-cope-called-monsters-1471237>

Pokémon stats (attack, defense, etc.) are given in `pokemon_stats.csv`. The mapping of stat numbers to names is given in `stats.csv`. Each Pokémon has an additive deviation from its species' base stats, known as IVs (*individual values*). IVs for each stat are uniformly distributed in the range [0,15]. Generate IVs for each spawned Pokémon for each of HP, attack, defense, speed, special attack, and special defense.³

The mechanics of Pokémon gender are somewhat involved (designed to reduce storage, see https://bulbapedia.bulbagarden.net/wiki/Individual_values). Some Pokémon don't even have genders. We'll simplify this by making all species have equal probability of being male or female. As with everything else here, you're welcome to implement the more complex mechanic if you like.

Pokémon have a 1/8192 chance of being shiny (a different color than normal), based on their IVs (see the above link). Again, simplify this by ignoring the IVs and simply using, e.g., `rand() % 8192 == 0`. Feel free to reduce that modulus to increase your shiny rate.

In order to level up your Pokémon, we'll apply the following formulae:

$$HP = \left\lfloor \frac{((base + IV) \times 2) \times level}{100} \right\rfloor + level + 10$$
$$Otherstat = \left\lfloor \frac{((base + IV) \times 2) \times level}{100} \right\rfloor + 5$$

where *base* is the Pokémon species' base stat, *IV* is its IV for the given stat and *level* is the Pokémon's level. Again, we're simplifying an MSG mechanic here by ignoring EVs (*effort values*), which normally add an additional boost to Pokémon growth. See <https://bulbapedia.bulbagarden.net/wiki/Stat> for the details.

When encountering Pokémon, we'll add a placeholder function for the battle and capture sequence in which we'll print the Pokémon's name, level, stats, and moves and pause for input (escape, any key, etc.).

In order to battle (next week), a trainer must have at least one Pokémon. Create an interface to give the PC the choice of three randomly generated level 1 Pokémon when the game starts. All generated trainers (NPCs) should be given at least one random Pokémon. Assuming the trainer has *n* Pokémon, there is a 60% probability that the trainer will get an (*n* + 1)th Pokémon, up to a maximum of 6 Pokémon. The generated trainers' Pokémon should be generated with levels and movesets as per the Pokémon generation rules for their map. You should be giving NPCs Pokémon this week, and the trainer-battle placeholder function should be updated to print enough details about the NPCs's Pokémon that you (and the TAs) can confirm that your Pokémon generation code is working.

A final note: There are probably only two of you (maybe three) who know less about Pokémon I do. Please let me know if I got something qualitatively wrong here, or if you think of a better simplification to MSG mechanics than I've proposed.

³Pokémon NerdsScholars: If I'm doing something wrong here, please let me know. I have a vague recollection of a past student telling me about IVs going to 32 and a different formula...